The mod p galois representation of an elliptic curve is irreducible at supersingular primes

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1 Introduction

The following theorem is useful in the Iwasawa theory of elliptic curves at supersingular primes, and possibly other topics I know less about

Theorem 1.1. *Let $E/\mathbb{Q}$ be an elliptic curve and p a rational prime such that E has good supersingular reduction at p. Then $E[p]$ is irreducible as a $\mathrm{Gal}_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-representation. In fact, it is even irreducible as an I_p-representation, where I_p is the inertia group of some prime of $\overline{\mathbb{Q}}$ lying over p.*

There is a proof of this in the literature, [BDB07, Prop. 4.4]. It makes use of an isomorphism between the formal group of a supersingular elliptic curve and the lubin-tate group of $W(\mathbb{F}_{p^2})$, see for example in [Kob03, Sec. 8]. Alternatively, a generalization about supersingular modular forms is proven in [Edi92, Thm 2.6], and implies the fact about elliptic curves by modularity. This proof uses finite flat group schemes, which we will use. However, working on the automorphic side actually significantly increases the difficulty of their proof, since they need to use the structure of modular curves as $\mathbb{Z}$-schemes. Obviously this is worth it for the most general result, however if you are just seeking a proof for elliptic curves, one can significantly simplify their proof.

One difficulty with their argument is that they use the work of Raynaud [Ray74]. I do not know how to remove this from the proof. One particular sticking point is that they use proposition 3.3.2(3). This is a less well-known result of that paper, which holds in the case $e = p - 1$ (e is absolute ramification index of the base ring) instead of $e < p - 1$. As far as I am aware, there is not an english proof of this proposition, unlike much of the other important results of Raynaud, which can be found in for example [Tat97]. So another thing I would like to do is provide an english proof of this lemma.

This means my goal for this paper is to provide a reasonably simple alternative proof Theorem 1.1 using finite flat group schemes that does not require the reader to read any French sources. I apologize if I have missed an English explanation that already exists. My literature searching skills leave a lot to be desired.

In terms of prerequisites, I freely use the theory of elliptic curves at the level of [Sil09]. The more difficult requirement is familiarity with finite flat group schemes. I will use [Tat97] for this, and will require essentially its entire content. Finally, we need the existence of Néron models, but only to know that $E[p]/\mathbb{Z}$ is a quasifinite flat group scheme, is finite once you invert the primes of bad reduction, and is its own Cartier dual.

2 Finite flat group schemes

We will need some basic results on the algebra of finite flat group schemes that are not covered in [Tat97], and we will also need a more advanced result, [Ray74, Prop. 3.3.2(3)]. We will discuss both in this section.

2.1 Connected, étale, and local-local parts

We work over a mixed characteristic Henselian local ring $(R, \mathfrak{m})$ that is $\mathfrak{m}$-adically complete with fraction field K and residue field κ. Let G/R be a finite flat group scheme with special fiber G_κ and generic fiber G_K. We can decompose G into three parts as follows. First, we have the connected-étale sequence[Tat97, 3.7 I]

$$0 \longrightarrow G^0 \longrightarrow G \longrightarrow G^{\text{ét}} \longrightarrow 0$$

Now, take G^0 and compute the connected-etale sequence of its Cartier dual

$$0 \longrightarrow ((G^0)^D)^0 \longrightarrow (G^0)^D \longrightarrow ((G^0)^D)^{\text{ét}} \longrightarrow 0$$

Dualizing this again, we obtain a decomposition of G^0 into $G^{\text{m}} \coloneqq (((G^0)^D)^{\text{ét}})^D$, called the multiplicative part, and $G^{\text{l}} \coloneqq (((G^0)^D)^0)^D$, called the local-local part.

If κ is a perfect field, then all of these sequences split via the section $G^{\text{ét}} \to G$ given by G^{red}. This means in this case we can write our decomposition as

$$G \cong (G^{\text{m}} \rtimes G^{\text{l}}) \rtimes G^{\text{ét}}.$$

If G is commutative, these semidirect products are products.

Remark. Note that while in theory we could also apply the connected-étale sequence to the Cartier dual of $G^{\text{ét}}$ to obtain a decomposition into four parts, this is not really necessary in our case. In fact, for p-groups, this will not change the decomposition at all, since the Cartier dual of an étale p-group is entirely connected. This follows from the fact that $\underline{\mathbb{Z}/p\mathbb{Z}}^D = \mu_p$ and a Dévissage argument (perhaps using Lemma 2.3).

Putting together this discussion, we obtain

Proposition 2.1. *Let G/R be a commutative finite flat p-group scheme over a mixed characteristic Henselian DVR with perfect field of fractions. Then there is a canonical decomposition*

$$G \cong G^m \times G^l \times G^{ét}$$

where G^m and G^l are connected and $G^{ét}$ is étale. Cartier duality interchanges the étale part and the multiplicative part, and fixes the local-local part.

Next we have an alternative classification of the connected component of a group scheme.

Lemma 2.2. *Let G/κ be a group scheme. let Fr denote the absolute Frobenius and let $G[Fr^n]$ denote the kernel of Fr^n. Then $G[Fr^n]$ stabalizes for large n, and its limit is G^0.*

Proof. Suppose first that G is connected and let A be its Hopf κ-algebra. Then it follows by CRT and the fact that $\operatorname{Spec}(A)$ is connected that

$$A \cong \kappa[X_1, \dots, X_n]/(X_1^{e_1}, \dots, X_k^{e_k}).$$

Then it is clear that for $n > \max\{\log_p(e_i) | 1 \le i \le k\}$, the following diagram commutes

$$\begin{tikzcd}
A \arrow[rr, "\mathrm{Fr}^n"] \arrow[dr, "e"'] & & A \\
& K \arrow[ur] &
\end{tikzcd}$$

where e is the counit, since $\mathrm{Fr}^n(X_1) = X_1^{p^n} \equiv 0 \bmod X_1^{e_1}$. But the fact that this diagram commutes is exactly the same as G being killed by Fr^n, since $G \to \operatorname{Spec} \kappa \to G$ is the 0 map.

Now let G be arbitrary. We need to show that $G[\mathrm{Fr}^n] \subseteq G^0$ for all n, after which we will obtain the result. Choosing n large, we may assume that $G^0 \subseteq G[F^n]$ by the previous result, so everything factors through $G^{\text{ét}}$ and we can assume that G is étale. So

$$G \cong \bigsqcup_i \operatorname{Spec} \kappa_i$$

as schemes, where κ_i are finite separable extensions of κ. Then it is clear that the Frobenius is an automorphism of G, so the kernel here is trivial.

Finally, we have one more lemma

Lemma 2.3. *Suppose G/S is a finite flat group scheme and $\varphi : S' \to S$ is faithfully flat. Then $G_{S'}$ is étale if and only if G_S itself is.*

Proof. If G_S is étale, then $G_{S'}$ is etale since etaleness is stable under arbitrary base change. Suppose G_S is not étale. Then since $G_S \to S$ is flat, it must be ramified, which since it is finite (and therefore finitely presented) that means we must have $\Omega_{G/S} \neq 0$. Then $\omega_{G_{S'}/S'} \cong \Omega_{G/S} \otimes S' \neq 0$ since faithfully flat morphisms preserve nontriviality of modules. □

□

2.2 Raynaud when $e = p - 1$

Now we will need to use a result of Raynaud about finite flat group schemes of order p^n when $e := v(p) = p-1$. Recall that if G_0/K is a finite flat group scheme, a prolongation of G_0 is a group scheme G/R such that $G_K \cong G_0$. Raynaud's most famous result is that if R is a mixed characteristic Henselian DVR with absolute ramification index $e < p - 1$, and G_0/K is a p-power order finite flat group scheme, then G_0 has at most one prolongation to R. We will need a version of this result for $e = p - 1$, but before that we need a lemma and some setup

Suppose that R is strictly Henselian. If F is a finite field, then by a Raynaud F-module (scheme) over R we mean a group scheme G/R with a map $F \to \mathrm{End}(G)$. By [Tat97, 4.4.1], if such a G is simple, it is represented by a Hopf algebra A generated by elements X_i, for $i \in \mathcal{I}$, where $\mathcal{I}$ is a principal homogeneous space of $\mathrm{Gal}(F/\mathbb{F}_p) \cong \mathbb{Z}/n\mathbb{Z}$. The relations in A are $X_i^p = \delta_i X_{i+1}$, where $\delta_i \in R$ obeys $0 \le v(\delta_i) \le e$.

If G/R and G'/R are two simple Raynaud F-modules with coefficients δ_i and δ_i', then a homomorphism $G' \to G$ that is an isomorphism on the generic fiber is determined by elements $a_i \in R$ such that

$$a_{i+1}\delta_i = a_i^p \delta_i'.$$

The map on Hopf algebras is given by $X_i \mapsto a_i X_i'$.

Lemma 2.4. *Let G and G' be two simple Raynaud F-modules over a mixed characteristic strictly Henselian DVR R with $e = p - 1$, and let $u : G' \to G$ be a homomorphism that is an isomorphism on generic fibers. Then either $v(a_i) = 0$ for all i, in which case $G \cong G'$ or $v(a_i) = 1$ for all i, in which case $v(\delta_i) = 0$ and $v(\delta_i') = p - 1$.*

Proof. By the equation $a_{i+1}\delta_i = a_i^p \delta_i'$, we get $pv(a_i) - v(a_{i+1}) = v(\delta_i) - v(\delta_i')$. The proof will proceed by abusing this formula.

If there is some i such that $v(a_i) > v(a_{i+1})$, then we get $p - 1 < pv(a_i) - v(a_{i+1}) = v(\delta_i) - v(\delta_i') \le p - 1$, which is a contradiction. Therefore $v(a_i)$ is constant.

If $v(a_i) = 0$, then $X_i \mapsto a_i X_i'$ is invertible since a_i is invertible, so we have that u is an isomorphism. If $v(a_i) > 0$ then we obtain $(p-1)v(a_i) = pv(a_i) - v(a_{i+1}) \le v(\delta_i) - v(\delta_i') \le p - 1$, so $v(a_i) = 1$, $v(\delta_i) = p - 1$, and $v(\delta_i') = 0$. □

Theorem 2.5 ([Ray74], Proposition 3.3.2(3))**.** *Suppose R is a mixed characteristic (not necessarily strictly) Henselian DVR of absolute ramification index $e = p - 1$. Let G_0/K be a finite flat group scheme killed by p that is simple over the strict henselization of R. Then G_0 has has at most two prolongations to R. If it has two prolongations, say G^+ and G^- with $G^+ > G^-$, then G^+ is étale and G^- is multiplicative.*

Proof. First assume R is strictly Henselian. Since G_0 is simple and killed by p, it is a Raynaud F-module, and therefore its maximal prolongation G^+ and minimal prolongation G^- are also Raynaud F-module schemes. Let A^+ and A^- be their associated Hopf algebras. By Lemma 2.4, we get either $G^+ = G^-$, in which case we are done, or if $G^+ \neq G^-$ we have $v(\delta_i^+) = 0$ and $v(\delta_i^-) = p - 1$, with the obvious notation.

Next we will show that in this case G^+ is étale and G^- is multiplicative. In fact, we only need to show that G^+ is étale, and G^- will follow by Cartier duality. As a ring, we have

$$A^+ = R[X_i^+ : i \in \mathcal{I}]/((X_i^+)^p - \delta_i^+ X_{i+1}^+ : i \in \mathcal{I})$$

Since A^+ is a Group scheme, it is étale if and only if it is étale at the identity section, which corresponds to $X_i^+ = 0$. Choosing an identification of $\mathcal{I}$ with $\mathbb{Z}/n\mathbb{Z}$, we let $f_i = (X_i^+)^p - \delta_i^+ X_{i+1}^+$, so

$$\mathrm{Jac}(f)|_{X_i^+=0} = \begin{pmatrix} 0 & -\delta_1^+ & 0 & \cdots & 0 \\ 0 & 0 & -\delta_2^+ & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -\delta_0^+ & 0 & 0 & \cdots & 0 \end{pmatrix}$$

and $\det(\mathrm{Jac}(f)|_{X_i^+=0}) = -\prod_{i\in\mathcal{I}} \delta_i \in R^\times$. Therefore A^+ is étale at 0 and therefore étale.

Since A^+ is étale, we know that $K(G_0)/K$ is an unramified extension. But by assumption R is strictly Henselian, so therefore $K = K(G_0)$ and G_0 is in fact the constant group scheme $\underline{\mathbb{Z}/p\mathbb{Z}}$, and $F = \mathbb{F}_p$. But if G is a group scheme killed by p, it is automatically an $\mathbb{F}_p$ module scheme. Therefore not only are G^+ and G^- Raynaud F-module schemes, but any prolongation G of G_0 is an F-module scheme. By Lemma 2.4, applied to the maps $G^+ \to G$ and $G \to G^-$, it follows that $G = G^+$ or $G = G^-$.

Now assume that R is not strictly Henselian and only regular Henselian. Let $\hat{R}$ be its strict Henselization, and suppose $G_0 \leq G_1 \leq G_2$ are prolongations of R. If any of these inequalities are strict, then their strictness is preserved under base change to $\hat{R}$, since $\hat{R}$ is a faithfully flat R-module and the strictness of these inequalities is just a fact about rings being proper subrings, which is stable under faithfully flat base change. Therefore the general R case follows from the Henselian case. ☐

Remark. We will only need the strictly Henselian case.

3 The proof

The proof will come in two steps. First, assuming E has supersingular reduction at p, we will determine some facts about the group scheme $E[p]_{\mathbb{Z}_p}$. Then we will assume that $E[p]$ is reducible, and determine different facts about $E[p]_{\mathbb{Z}_p}$. Then we will observe that these facts are contradictory.

Proof of Theorem 1.1. First assume E is supersingular at p. We claim that $E[p]_{\mathbb{Z}_p}$ is connected. To see this, first reduce E mod p to obtain $E[p]_{\mathbb{F}_p}$. The characteristic polynomial of the frobenius ϕ is $x^2 - a_p x + p \equiv x^2 \bmod p$, since $a_p \equiv 0 \bmod p$ is an equivalent condition for supersingularity. This implies that $\phi^2 = 0$ on $E[p]_{\mathbb{F}_p}$, so by Lemma 2.2, it follows that $E[p]_{\mathbb{F}_p}$ and therefore $E[p]_{\mathbb{Z}_p}$ is connected.

Now assume that $E[p]_{\mathbb{Q}}$ is reducible as a I_p-module, where I_p is some inertia group of a prime of $\overline{\mathbb{Q}}$ lying over p (Note that this would follow from $E[p]_{\mathbb{Q}}$ being reducible as a $\mathrm{Gal}_{\mathbb{Q}}$-module). It therefore fits into a short exact sequence of I_p-modules:

$$0 \longrightarrow A \longrightarrow E[p] \longrightarrow B \longrightarrow 0$$

Choosing a basis v_1, v_2 of $E[p]_{\mathbb{Q}_p}$ with $v \in A$, we see that the Galois representation $\rho : I_p \to \mathrm{GL}_2(\mathbb{F}_p)$ defined by $E[p]$ takes the shape

$$\rho(g) = \begin{pmatrix} \chi(g) & b(g) \\ 0 & \psi(g) \end{pmatrix}$$

where $\chi(g)$ and $\psi(g)$ are the characters $I_p \to \mathbb{F}_p^\times$ corresponding to A and B. Let $K = \hat{\mathbb{Q}}_p^{\mathrm{unr}}$ and $L = \hat{\mathbb{Q}}_p^{\mathrm{unr}}(A)$ be the smallest extension of K where A is a trivial G_K-module. Then $[L : K] \leq p-1$ because $\mathbb{F}_p^\times$ has cardinality $p-1$. In particular, the absolute ramification index of L is at most $p-1$.

If the ramification index less than $p-1$, then A_L has at most one prolongation to $\mathcal{O}_L$ by [Tat97, Thm 4.5.1]. Since $A_L \cong \underline{\mathbb{Z}/p\mathbb{Z}}_L$, one such prolongation is $\underline{\mathbb{Z}/p\mathbb{Z}}_{\mathcal{O}_L}$. In particular, the closure of A_L in $E[p]_{\mathcal{O}_L}$ is $\underline{\mathbb{Z}/p\mathbb{Z}}_{\mathcal{O}_L}$, and therefore étale. It follows then that the closure of A_K in $E[p]_{\mathcal{O}_K}$, call that $A_{\mathcal{O}_K}$, is also étale by Lemma 2.3. So $E[p]_{\mathbb{Z}_p}$ has a nontrivial étale part in this case.

If the inertial index is equal to $p-1$, then A_L has at most two prolongations to $\mathcal{O}_L$ by Theorem 2.5. One such prolongation, for the same reasons as in the previous problem, is $\underline{\mathbb{Z}/p\mathbb{Z}}_{\mathcal{O}_L}$. If there are two, then the other must be of multiplicative type. In any case, the closure of A_L in $E[p]_{\mathcal{O}_K}$ is a prolongation of A_L,

$A_{\mathcal{O}_L}$, that is either étale or multiplicative. Therefore either $C = A$ or $C = A^D$ must satisfy $C_{\mathcal{O}_L}$ is étale by Proposition 2.1. It follows again by 2.3 that C is étale, so since $E[p] = E[p]^D$, C $E[p]_{\mathbb{Z}_p}$ once again has a nontrivial étale part.

We have proven that if p is supersingular, then $E[p]$ is connected. But if $E[p]$ is reducible as an I_p-module, then E has an étale part. By Proposition 2.1, these cases are mutually exclusive. □

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